

General Boolean Spectral Maxima and the Missing Type-I and Type-III Extreme Values

Autonomous solution packet for arXiv:2003.05382, Problem 2.17

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Abstract

We answer both clauses of Problem 2.17. For any two probability laws on $\mathbb{R}$, the canonical Boolean product gives Boolean-independent selfadjoint operators with those laws whose spectral maximum has an explicit distribution. At negative thresholds its CDF is an endpoint test; at nonnegative thresholds it is $FG/(F+G-FG)$. This defines an associative max-operation on all real laws. The same model, followed by ordinary affine normalization of the maximum, realizes three Boolean extreme-value limits: Dagum (type II), logistic (type I), and a bounded reciprocal-power family (type III).

1 The question and the answer

Problem 2.17 of Ueda asks for operator models realizing the maximum of Boolean-independent general selfadjoint operators and for Boolean extreme values other than the positive Dagum family [1]. The exact source text is reproduced below.

Problem 2.17. We do not know operator models which realize the maximum of Boolean independent general selfadjoint operators. Can we obtain such operator models? Moreover, we should find other type Boolean extreme values.

For $p, q \in [0, 1]$, put

$$p \mathbin{\lrcorner}_B q = \begin{cases} \frac{pq}{p+q-pq}, & p+q > 0, \\ 0, & p=q=0. \end{cases} \quad (1)$$

Thus $(p \mathbin{\smash{\,\,\mathsf{\tiny \wedge}}\,}_B q)^{-1} - 1 = (p^{-1} - 1) + (q^{-1} - 1)$ when $p, q > 0$. This is the scalar Boolean projection operation of Vargas–Voiculescu [2, Lemma 3.1].

Theorem 1 (arbitrary-selfadjoint Boolean maximum). *Let μ_1, μ_2 be arbitrary probability measures on $\mathbb{R}$. There are Boolean-independent, possibly unbounded, selfadjoint operators $\hat{X}_1, \hat{X}_2$ with respective vector-state laws μ_1, μ_2 . If $F_i(t) = \mu_i((-\infty, t])$, then their spectral maximum has CDF*

$$F_{\widehat{X}_1 \vee \widehat{X}_2}(t) = \begin{cases} 1, & t < 0 \text{ and } F_1(t) = F_2(t) = 1, \\ 0, & t < 0 \text{ and at least one of } F_1(t), F_2(t) \text{ is } < 1, \\ F_1(t) \wedge_B F_2(t), & t \geq 0. \end{cases} \quad (2)$$

The construction is canonical from the two marginal multiplication models. Formula (2) defines an associative operation on all real distribution functions and is realized simultaneously for every finite number of Boolean-independent copies.

The distinguished point 0 is not an artificial choice: it is the zero operator on the orthogonal complement introduced by the nonunital Boolean product. Positivity in the earlier theory merely ensures that only the third line of (2) is visible.

Theorem 2 (three Boolean extreme-value types). *For every $\alpha > 0$, normalized spectral maxima of positive, Boolean-independent selfadjoint operators in the model of Theorem 1 have each of the following nondegenerate limits:*

$$\begin{aligned} B_{\text{II},\alpha}(x) &= \begin{cases} (1+x^{-\alpha})^{-1}, & x > 0, \\ 0, & x \leq 0, \end{cases} \\ B_{\text{I}}(x) &= (1+e^{-x})^{-1}, \quad x \in \mathbb{R}, \\ B_{\text{III},\alpha}(x) &= \begin{cases} (1+(-x)^\alpha)^{-1}, & x \leq 0, \\ 1, & x > 0. \end{cases} \end{aligned}$$

The first is the known Dagum/type-II law. The logistic law B_{I} and the bounded law $B_{\text{III},\alpha}$ give the requested type-I and type-III Boolean extreme values.

2 Proof intuition

The Boolean product Hilbert space contains the two centered marginal spaces and one shared vacuum line. Each variable acts as its original multiplication operator on its own centered space plus the vacuum, and as zero on the other centered space. Consequently its lower spectral projection has two different forms. Below zero the foreign zero-space is excluded; at and above zero it is included. Intersecting the two lower spectral subspaces therefore gives an endpoint test below zero and the familiar Boolean projection formula above zero.

For extreme values, the scalar homeomorphism

$$\chi(u) = \begin{cases} e^{1-u^{-1}}, & u > 0, \\ 0, & u = 0, \end{cases} \quad \chi^{-1}(v) = \begin{cases} (1 - \log v)^{-1}, & v > 0, \\ 0, & v = 0, \end{cases}$$

turns $\wedge_B$ into multiplication. Thus positive Boolean maxima are the pullback under χ of classical maxima. Allowing the usual centering only *after* taking the operator maximum avoids the unit problem noted in the earlier Boolean theory. Pulling back the three classical limits gives the three formulas in Theorem 2.

3 The canonical operator model

For $i = 1, 2$, let

$$H_i = L^2(\mathbb{R}, \mu_i), \quad \xi_i = \mathbf{1}, \quad X_i = M_x,$$

where M_x is the maximal multiplication operator. Put $K_i = H_i \ominus \mathbb{C}\xi_i$ and

$$H = \mathbb{C}\Omega \oplus K_1 \oplus K_2.$$

Let $V_i : H_i \rightarrow H$ be the isometry that sends ξ_i to Ω and is the identity from K_i to its displayed summand. Define

$$\widehat{X}_1 = (V_1 X_1 V_1^*) \oplus 0_{K_2}, \quad \widehat{X}_2 = (V_2 X_2 V_2^*) \oplus 0_{K_1}, \quad (3)$$

with the evident direct-sum domains. These are selfadjoint even when the marginal multiplication operators are unbounded, and $\langle f(\widehat{X}_i)\Omega, \Omega \rangle = \int f d\mu_i$.

The nonunital operator families $V_i \mathcal{B}(H_i) V_i^*$ are Boolean independent for the vacuum state. Indeed, if adjacent indices alternate, then $V_i^* V_j = |\xi_i\rangle \langle \xi_j|$ for $i \neq j$, so direct multiplication gives

$$\langle A_1 \cdots A_m \Omega, \Omega \rangle = \prod_{k=1}^m \langle A_k \Omega, \Omega \rangle.$$

This is precisely Boolean factorization. It also proves that (3) depends only on the two marginal models.

We record the projection computation used below.

Lemma 3 (Boolean projection join). *Let R, S be Boolean-independent projections and set $p = \varphi(I - R)$, $q = \varphi(I - S)$. Then*

$$\varphi(I - (R \vee S)) = p \wedge_B q.$$

Proof. This is [2, Lemma 3.1]; a short derivation makes the packet self-contained. The Cauchy and Boolean cumulant transforms of R are

$$G_R(z) = \frac{p}{z} + \frac{1-p}{z-1} = \frac{z-p}{z(z-1)}, \quad K_R(z) = z - G_R(z)^{-1} = \frac{z(1-p)}{z-p},$$

and similarly for S . Boolean additivity gives $K_{R+S} = K_R + K_S$. Since $\ker(R + S) = \ker R \cap \ker S = \text{ran}(I - (R \vee S))$, its vacuum mass is the atom of $R + S$ at zero. Hence

$$\varphi(I - (R \vee S)) = \lim_{z \rightarrow 0} z G_{R+S}(z) = \frac{1}{p^{-1} + q^{-1} - 1} = p \wedge_B q,$$

with the endpoint cases obtained by continuity. $\square$

Proof of Theorem 1. Let $A_i(t) = E_{X_i}((-\infty, t])$ and $P_i(t) = E_{\widehat{X}_i}((-\infty, t])$. Functional calculus in the direct sum (3) gives

$$P_1(t) = \begin{cases} V_1 A_1(t) V_1^*, & t < 0, \\ V_1 A_1(t) V_1^* + P_{K_2}, & t \geq 0, \end{cases} \quad (4)$$

and the symmetric formula for $P_2(t)$.

Suppose first that $t < 0$. The ranges of $P_1(t)$ and $P_2(t)$ lie in $\mathbb{C}\Omega \oplus K_1$ and $\mathbb{C}\Omega \oplus K_2$, whose intersection is $\mathbb{C}\Omega$. Their intersection contains Ω if and only if $A_i(t)\xi_i = \xi_i$ for both i . Since $A_i(t)$ is a projection, this is equivalent to $\langle A_i(t)\xi_i, \xi_i \rangle = F_i(t) = 1$. Thus the first two lines of (2) follow.

Now let $t \geq 0$. Formula (4) yields

$$I - P_i(t) = V_i(I - A_i(t))V_i^* =: R_i(t).$$

The two $R_i(t)$ are Boolean-independent projections and $\varphi(I - R_i(t)) = F_i(t)$. Because the lower spectral projection of the spectral maximum is $P_1(t) \wedge P_2(t)$, Lemma 3 gives

$$F_{\widehat{X}_1 \vee \widehat{X}_2}(t) = \varphi(P_1(t) \wedge P_2(t)) = \varphi(I - (R_1(t) \vee R_2(t))) = F_1(t) \wedge_B F_2(t).$$

The right side of (2) is increasing and right-continuous; at zero its left limit can be one only when both $F_i(0) = 1$. It therefore is a CDF. Below zero its scalar operation is “one iff every input is one,” which is associative. At nonnegative thresholds, associativity follows from the additivity of $u^{-1} - 1$. For finitely many marginal spaces, replace H by $\mathbb{C}\Omega \oplus \bigoplus_i K_i$. The same proof, or the atom at zero of the sum of all complementary projections, gives the corresponding finite-fold formula. $\square$

For a CDF F , write $F^{\diamond n}$ for the CDF of the maximum of n canonical Boolean copies. The preceding proof gives, for $n \geq 2$,

$$F^{\diamond n}(t) = \begin{cases} \mathbf{1}_{\{F(t)=1\}}, & t < 0, \\ \frac{F(t)}{n - (n-1)F(t)}, & t \geq 0. \end{cases} \quad (5)$$

4 Extreme-value transfer and the two new types

The identity

$$\chi(p \wedge_B q) = \chi(p)\chi(q) \quad (6)$$

is immediate from (1). Consequently, if K is a CDF supported on $[0, \infty)$ and $F = \chi^{-1} \circ K$, then, whenever $y \geq 0$,

$$\chi(F^{\diamond n}(y)) = K(y)^n. \quad (7)$$

Thus every classical normalization whose evaluation points are eventually nonnegative transfers to the Boolean model. We now give explicit parents, so no appeal to a general domain-of-attraction theorem is needed.

Type II (Fréchet $\rightarrow$ Dagum). Let

$$K_{0,\text{II}}(y) = \begin{cases} 0, & y < 1, \\ 1 - y^{-\alpha}, & y \geq 1, \end{cases} \quad F_{0,\text{II}} = \chi^{-1} \circ K_{0,\text{II}}.$$

If M_n is the spectral maximum of n canonical Boolean copies with law $F_{0,\text{II}}$, then, for $x > 0$,

$$\chi(F_{M_n/n^{1/\alpha}}(x)) = \left(1 - \frac{x^{-\alpha}}{n}\right)^n \rightarrow e^{-x^{-\alpha}}.$$

For $x \leq 0$ the limit is zero. Applying χ^{-1} gives $B_{\text{II},\alpha}$.

Type I (Gumbel $\rightarrow$ logistic). Take the exponential CDF

$$K_{0,\text{I}}(y) = \begin{cases} 0, & y < 0, \\ 1 - e^{-y}, & y \geq 0, \end{cases} \quad F_{0,\text{I}} = \chi^{-1} \circ K_{0,\text{I}}.$$

For every fixed x , the point $x + \log n$ is nonnegative for all large n , and (7) gives

$$\chi(F_{M_n - \log n}(x)) = \left(1 - \frac{e^{-x}}{n}\right)^n \rightarrow e^{-e^{-x}}.$$

Therefore $M_n - \log n$ converges in distribution to

$$\chi^{-1}(e^{-e^{-x}}) = \frac{1}{1 + e^{-x}},$$

the logistic CDF.

Type III (reverse Weibull $\rightarrow$ bounded reciprocal power). Let

$$K_{0,\text{III}}(y) = \begin{cases} 0, & y < 0, \\ 1 - (1 - y)^\alpha, & 0 \leq y \leq 1, \\ 1, & y > 1, \end{cases} \quad F_{0,\text{III}} = \chi^{-1} \circ K_{0,\text{III}}.$$

For $x \leq 0$ and all large n ,

$$\chi(F_{n^{1/\alpha}(M_n - 1)}(x)) = \left(1 - \frac{(-x)^\alpha}{n}\right)^n \rightarrow e^{-(-x)^\alpha};$$

for $x > 0$ the CDF is already one. Applying χ^{-1} gives $B_{\text{III},\alpha}$. This proves Theorem 2.

5 Why centering is legitimate

Earlier treatments avoided shifts of the *inputs*, because adjoining a common scalar identity to Boolean-independent nonunital algebras destroys the naive Boolean product picture [2, Section 2]. Here no input is shifted. We first form the spectral maximum M_n inside the canonical Boolean product and only then apply the ordinary selfadjoint affine map $a^{-1}(M_n - bI)$. For $a > 0$ its CDF is exactly $F_{M_n}(ax + b)$, which is all that was used above. Hence the type-I and type-III constructions do not assume Boolean independence of translated copies and do not encounter the unit obstruction.

6 Scope and verification

Theorem 1 is an existence model determined by the marginals; it does not assert that the spectral maximum has the same law in every possible realization of Boolean independence. This is precisely the operator-model existence requested in Problem 2.17. The formulas include atoms, unbounded marginal multiplication operators, and all threshold edge cases.

The accompanying verifier checks (6), the type-I and type-III limits, and finite-dimensional canonical models with mixed-sign and purely negative spectra. In those models it constructs the lower spectral projections, computes their range intersection numerically, and compares the vacuum mass to (2).

References

- [1] Y. Ueda, *Max-convolution semigroups and extreme values in limit theorems for the free multiplicative convolution*, arXiv:2003.05382 (2020), Problem 2.17.
- [2] J. G. Vargas and D. V. Voiculescu, *Boolean Extremes and Dagum Distributions*, arXiv:1711.06227 (2018), especially Sections 2–4 and Lemmas 3.1–3.3.